

Attention-Guided Graph Neural Networks for Protein Pathway Prediction and Functional Clustering

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Abstract—Proteins seldom act in isolation. They interact through various sensitive networks that give rise to cellular pathways. Having a fundamental understanding of how proteins interact with one another is of paramount importance to combat various disease mechanisms and can help in accelerating pharmaceutical development. It has been noticed, however, that traditional mechanisms fall short on being able to accurately model and classify these networks. Additionally, they fail to solve the issue of their inherent uncertainty. We propose an attention-guided Graph Neural Network (GNN) framework designed to handle the intricacies of the relationships between different protein structures. We address the uncertainty problem by integrating the Monte Carlo dropout method. This approach offers confidence-based embeddings to measure the uncertainty of prediction. We use attention-guided message transmitting to adaptively prioritize biologically significant protein interactions. We used clustering on the learned embeddings to reconstruct pathways and uncover higher-order biological organization. These pathways are then verified by enrichment analysis against curated KEGG pathways. In doing this, our framework provides accuracy and interpretability by addressing pathway prediction and functional clustering together.

Index Terms—Protein-protein interaction (PPI), graph neural networks (GNNs), attention mechanisms, Monte Carlo dropout, pathway prediction, bioinformatics

I. INTRODUCTION

Proteins form extensive networks of interactions in the form of protein-protein interactions (PPIs), which drive cellular pathways and regulate essential biological functions [5]. Deciphering these pathways is crucial to understanding the molecular underpinnings of health and disease because disruptions in their organisation are frequently connected to pathological conditions. Precise pathway prediction has the potential to accelerate pharmaceutical innovation, foster the identification of therapeutic targets, and advance biological knowledge at the systems level [4]. Because large-scale protein-protein interaction (PPI) datasets are growing exponentially, computational methods are now crucial for reconstructing pathways and exposing higher-order biological organisation.

With their success in citation networks, social graphs, and biological networks, graph-based learning techniques have recently become highly effective tools for modelling relational

data [1], [3]. Beyond conventional machine learning techniques that depend on manually created features, graph neural networks (GNNs) show real promise in the bioinformatics field for capturing the non-Euclidean structure of PPI data. Recent developments have made it possible for models to prioritize informative neighbours and enhance interpretability, such as the addition of attention mechanisms to GNN architectures [1]. Similar architectures and pooling strategies have also been used by domain-specific frameworks such as ASCE-PPIS to achieve high performance in protein interaction site prediction [2]. Protein pathway clustering and prediction are still difficult tasks in spite of these developments.

Current approaches have the following drawbacks:

- They frequently treat all interactions equally despite their differing biological significance. Thus, they fail to accurately represent the heterogeneity of PPI networks.
- Prediction uncertainty is rarely addressed, which lowers dependability in biomedical applications that follow. This also hinders its applicability in medical domains.
- Although node- or edge-level tasks are the focus of current frameworks, higher-level clustering is rarely integrated to reconstruct coherent biological pathways.

To address these challenges, we propose an attention-guided Graph Neural Network (GNN) framework that addresses the uncertainty problem through a Bayesian approximation technique called the Monte Carlo dropout. The attention mechanism is a key factor that enables the dynamic weighting of protein interactions, thus it can easily find biologically relevant neighbours for message passing. With the integration of Monte Carlo dropout, the system is able to estimate the uncertainty in the embeddings, offering insight into prediction stability. Clustering methods are applied on these embeddings to perform the reconstruction of the pathways that are then verified by enrichment analysis comparing with the annotated KEGG pathways. This unified approach directly tackles both pathway prediction and functional clustering.

This work focuses on three main contributions. First, we create an attention-guided GNN with a focus on adaptive neighbour weighting that is specifically designed for path-

way discovery in protein networks. Second, to reconstruct biologically significant pathways and validate them against outside sources, we integrate pathway enrichment analysis with clustering. Lastly, by incorporating Monte Carlo dropout for confidence estimation, we give uncertainty prediction top priority in pathway prediction. The potential of attention-driven, uncertainty-aware GNNs as dependable computational tools for protein pathway analysis and discovery is highlighted by these contributions taken together.

II. LITERATURE REVIEW

Graph-based learning has gained a lot of attention as a method for modeling complex relational data in a number of domains, including social networks, recommendation systems, and biological systems. This advancement has been largely fuelled by the development of graph neural networks (GNNs), which provide frameworks for applying deep learning to non-Euclidean data [3]. In order to enable expressive representations of graph-structured data, GNNs use iterative message-passing schemes to aggregate information from nearby nodes. Numerous GNN architecture families, including recurrent, convolutional, autoencoder-based, and spatial-temporal models, have been proposed over time [3]. Although these methods have proven effective in a wide range of application areas, the size, diversity, and unpredictability of protein–protein interaction (PPI) data present particular difficulties for their implementation in biological networks.

The addition of attention mechanisms, which enable models to determine the relative importance of nearby nodes during message passing, is a noteworthy development in GNN architecture. By giving neighbours adaptive weights, the Graph Attention Network (GAT), first presented by Veličković et al. [1], was one of the first to show how attention could enhance node-level classification and get around the drawbacks of uniform aggregation in graph convolutional networks. Performance on benchmark tasks, such as citation networks and inductive settings like protein–protein interaction datasets, was enhanced by this innovation. Even though attention enhances interpretability and task performance, the original design of GAT and related models did not specifically address prediction uncertainty, which is crucial in biological applications where dependability is crucial.

Simultaneously, scientists have increasingly sophisticatedly used GNNs for protein interaction tasks. To predict protein–protein interaction sites, the recently proposed ASCE-PPIS framework combines equivariant GNNs with structure-aware pooling and graph collapse [2]. ASCE-PPIS demonstrated the advantages of combining structural awareness with sophisticated graph architectures by outperforming previous methods on benchmark datasets by modelling geometric features and fusing protein large language model embeddings. Despite its effectiveness, ASCE-PPIS and related techniques are mainly limited to predicting residue-level binding sites; they are not capable of reconstructing pathways or grouping proteins into higher-level biological systems. Additionally, these models’ reliability in translational or clinical contexts

is limited because, like many of their predecessors, they lack mechanisms to quantify uncertainty.

Systems biology has long aimed to combine diverse data sources to reconstruct cellular pathways, extending beyond GNN-based techniques. Traditional bioinformatics methods rely heavily on network integration, statistical enrichment, and graph analysis to identify pathways from large datasets. While these methods provide valuable biological insights and allow comparison with curated databases like KEGG, they depend on manually designed features and deterministic models. As a result, they are less adaptable to the variability of modern protein–protein interaction (PPI) networks and often struggle to scale with the growing size and complexity of biological data. Moreover, they lack the representational flexibility that deep learning approaches offer.

The application of graph learning to biological networks has shown steady progress in the literature, as evidenced by the development of attention-based GNNs [1], thorough surveys that establish taxonomies of graph models [3], domain-specific frameworks for protein interactions [2], and bioinformatics pipelines for pathway analysis [4]. Key gaps still exist, though: current approaches rarely integrate clustering or enrichment validation to reconstruct biologically coherent pathways; they rarely incorporate uncertainty modelling, which reduces confidence in their predictions; and they offer limited interpretability regarding functionally significant protein interactions. To move from isolated predictions to reliable, systems-level pathway discovery, these gaps must be filled.

III. METHODOLOGY

A. Data Acquisition and Preprocessing

The STRING database (version 12.0), which offers computationally predicted and experimentally verified relationships between human proteins, provided the protein–protein interaction (PPI) data that we used. We implemented a strict confidence threshold to guarantee dependability, eliminating low-confidence interactions by keeping only edges with a combined score of 700. High-confidence PPI relationships appropriate for further analysis are captured in the resultant dataset. We included network embeddings and protein annotations (preferred names, sizes, and functional labels) from the STRING repository in addition to interaction data. Compact vector representations of proteins that encode relational information are provided by these embeddings, which were initially obtained from deep unsupervised learning techniques. To create node-level features, we aligned these embeddings with protein identifiers.

There were several steps in the preprocessing pipeline. To produce a complete dataset with node attributes and edge attributes, the protein metadata was combined with embeddings and interaction edges. We then constructed a graph representation with proteins as nodes and their interactions as edges weighted by confidence scores. Subgraphs were shown for quality control after the largest connected component of the graph was removed to confirm connectivity. Finally, the graph was converted into the PyTorch Geometric (PyG) data

format to enable efficient processing in the resulting GNN framework.

B. Model Overview

We represent the network of protein-protein interactions as a graph

$$G = (V, E),$$

where V represents the set of proteins and E denotes their pairwise interactions. Each node $i \in V$ is associated with an input feature vector

$$x_i \in \mathbb{R}^F,$$

derived from protein embeddings and annotations. The adjacency matrix

$$A \in \{0, 1\}^{|V| \times |V|}$$

encodes the interaction structure, with edges weighted by the STRING combined score.

Our framework extends the Graph Attention Network (GAT) architecture, which adds masked self-attention mechanisms to GNNs' message-passing paradigm. GAT captures the heterogeneity of protein-protein interactions by allowing each node to assign adaptive weights to its neighbors, in contrast to Graph Convolutional Networks (GCNs), which aggregate neighbor features uniformly.

The transformation begins with a shared linear projection of node features:

$$h_i = Wx_i, \quad W \in \mathbb{R}^{F' \times F},$$

where F and F' denote the input and output feature dimensions, respectively.

For each edge $(i, j) \in E$, an attention coefficient is computed to quantify the importance of node j 's features to node i :

$$e_{ij} = a(h_i, h_j) = \text{LeakyReLU}(a^\top [h_i \parallel h_j]),$$

where $a \in \mathbb{R}^{2F'}$ is a learnable attention vector and $\parallel$ denotes concatenation.

To ensure comparability, the coefficients are normalized across the neighborhood N_i of node i using the softmax function:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in N_i} \exp(e_{ik})}.$$

The normalized coefficients α_{ij} serve as adaptive weights in the aggregation step, producing updated node features:

$$h'_i = \sigma \left(\sum_{j \in N_i} \alpha_{ij} Wx_j \right),$$

where $\sigma(\cdot)$ denotes a nonlinear activation function (e.g., ELU).

To enhance stability and expressive power, multi-head attention is employed. Specifically, K independent attention mechanisms are applied in parallel, and their outputs are concatenated:

$$h'_i = \parallel_{k=1}^K \sigma \left(\sum_{j \in N_i} \alpha_{ij}^{(k)} W^{(k)} x_j \right).$$

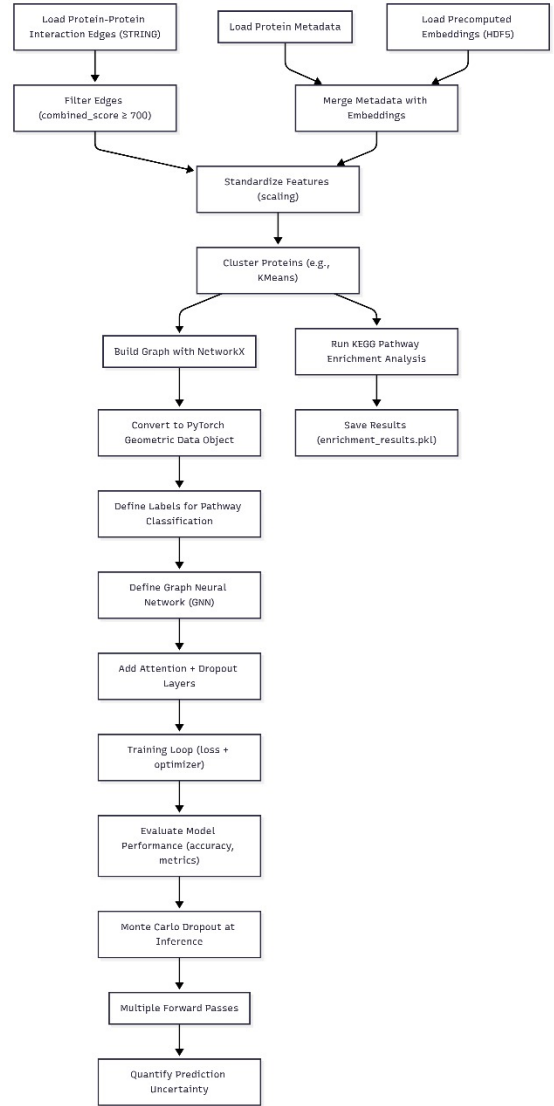


Fig. 1. Flowchart of the complete methodology, from data preprocessing to uncertainty quantification.

This formulation enables the model to attend to multiple complementary aspects of protein-protein interactions simultaneously.

By stacking multiple GAT layers, proteins are able to create higher-order representations by integrating contextual information from progressively larger neighbourhoods in the interaction graph. The final embeddings of each protein node, denoted by the symbol z_i , serve as the basis for further processes like pathway reconstruction and clustering.

To summarise, the GAT-based encoder provides a biologically motivated approach to adaptively weighing protein interactions, identifying functionally significant neighbours, and producing expressive node embeddings that form the foundation of the framework.

C. Embedding Generation with Graph Attention Networks

A multi-layer Graph Attention Network (GAT) encoder was fed the preprocessed protein–protein interaction graph. A feature vector was used to represent each protein node. It comprises of annotation metadata and embeddings derived from STRING database. The GAT model generated sophisticated node representations that incorporate evidence of local interactions and global network context through a series of attention-guided message-passing layers.

Formally, given the PPI graph

$$G = (V, E)$$

with node features

$$X \in \mathbb{R}^{N \times F},$$

the stacked GAT layers transform these features into a latent embedding matrix

$$Z = \text{GAT}(X, A) \in \mathbb{R}^{N \times d},$$

where A is the adjacency matrix, N is the number of proteins, and d is the dimension of the learned embeddings. Each row

$$\mathbf{z}_i \in \mathbb{R}^d$$

corresponds to the final representation of protein i , incorporating information from its neighbors according to the learned attention coefficients.

For tasks that follow, these embeddings offer a condensed and biologically informed representation of the PPI network. Specifically, they serve as the foundation for our uncertainty-aware learning approach, in which we create confidence-aware protein embeddings and quantify prediction stability using Monte Carlo dropout.

D. Clustering and Pathway Validation

Compact, biologically accurate representations of proteins are produced by the GAT encoder’s embeddings. We used clustering techniques on these embeddings to identify higher-order biological organisation by classifying proteins into putative pathways according to their learnt similarity. Because of its ease of use and efficiency in high-dimensional embedding spaces, KMeans was chosen as the main clustering algorithm among those examined.

Given the embedding matrix

$$Z \in \mathbb{R}^{N \times d},$$

the clustering objective is to partition proteins into K clusters by minimizing the within-cluster variance:

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{\mathbf{z}_i \in C_k} \|\mathbf{z}_i - \boldsymbol{\mu}_k\|^2,$$

where $\boldsymbol{\mu}_k$ denotes the centroid of cluster C_k . Apart from KMeans, we also used DBSCAN, a density-based technique that finds clusters of any shape while detecting outliers, and Spectral Clustering, which uses the eigen-decomposition of the

graph Laplacian to find community-like structures. These complementary methods ensured robustness in cluster formation by offering different viewpoints on how proteins are organised.

We used enrichment analysis against curated KEGG pathways to assess the reconstructed clusters’ biological significance. To determine whether each cluster included a statistically significant overlap with established pathways, we used a hypergeometric test. The probability of observing k or more overlapping genes between a cluster of size n and a pathway of size K , drawn from a background of N proteins, is given by:

$$p = \sum_{i=k}^n \frac{\binom{K}{i} \binom{N-K}{n-i}}{\binom{N}{n}}.$$

Clusters with low p -values exhibit strong correspondence to known biological pathways, whereas other clusters might represent novel or poorly understood functional modules. Our framework moves from individual protein embeddings to interpretable and biologically coherent pathway reconstructions, bridging the gap between representation learning and functional discovery.

E. Uncertainty Modeling with Monte Carlo Dropout

Although adaptively weighting neighbours improves the interpretability of protein-protein interactions, attention mechanisms do not specifically measure the degree of uncertainty in the learnt representations. In bioinformatics applications, where accurate predictions are necessary for biological discovery, this is a significant drawback. We use Monte Carlo (MC) dropout, which approximates Bayesian inference by enabling dropout layers during inference and aggregating the results across multiple stochastic forward passes, to address this problem.

Let $f_\theta(\cdot)$ denote the GAT encoder parameterized by θ , and let $\text{Dropout}(p)$ represent a stochastic masking operator with dropout probability p . Given the input graph data (X, A) , the t -th stochastic embedding of node i is expressed as

$$\mathbf{z}_i^{(t)} = f_\theta(X, A; \text{Dropout}(p)), \quad t = 1, 2, \dots, T$$

where T denotes the number of Monte Carlo samples. The collection of embeddings across all samples yields an empirical distribution

$$\{\mathbf{z}_i^{(1)}, \mathbf{z}_i^{(2)}, \dots, \mathbf{z}_i^{(T)}\}.$$

From this distribution, we compute the mean embedding as the confidence-aware representation of node i :

$$\bar{\mathbf{z}}_i = \frac{1}{T} \sum_{t=1}^T \mathbf{z}_i^{(t)},$$

and the variance as a measure of predictive uncertainty:

$$\text{Var}(\mathbf{z}_i) = \frac{1}{T} \sum_{t=1}^T (\mathbf{z}_i^{(t)} - \bar{\mathbf{z}}_i)^2.$$

Nodes with lower variance are linked to stable, high-confidence embeddings, whereas nodes with higher variance

indicate representational uncertainty. Using this uncertainty estimation, our framework generates expressive embeddings and gives each prediction a reliability score, providing robustness for further biological interpretation.

IV. IMPLEMENTATION

PyTorch Geometric was used to implement the suggested framework on top of PyTorch. Only high-confidence interactions with a combined score of at least 700 were kept when we built the protein–protein interaction (PPI) network using the STRING database (version 12.0). Interactions were encoded as weighted edges, and node-level features were created by combining protein metadata and precomputed embeddings from STRING. To ensure compatibility with training later on, the generated graph was processed into the PyTorch Geometric format. Both PyTorch and NumPy used fixed random seeds, and GPU acceleration was encouraged.

A Graph Attention Network (GAT) served as the main encoder. Neighbourhood interactions were adaptively weighted and projected through stacked attention layers for each protein node feature vector. The output layer generated a 16-dimensional embedding with hidden dimensions set to 64. Training was stabilised by multi-head attention, and dropout was used in both the training and evaluation stages. The Adam optimiser was used to train the model, and early stopping was determined by validation performance. Classification experiments used a standard cross-entropy loss, while link prediction experiments used negative edge sampling and edge splits from PyTorch Geometric.

After training, pathways were reconstructed using unsupervised clustering using the learnt embeddings. We mostly used KMeans clustering, but for robustness checks, we also used Spectral Clustering and DBSCAN. Proteins were clustered into higher-order modules, which were then confirmed by enrichment analysis using carefully chosen KEGG pathways. A hypergeometric test, which quantifies the statistical overlap between reconstructed clusters and known pathway gene sets, was used to evaluate enrichment. While some clusters were assessed for possible novel associations, others that displayed significant enrichment values were considered biologically significant.

Lastly, we used Monte Carlo dropout to evaluate the degree of uncertainty in the cluster assignments and learnt embeddings. The framework produced distributions over protein embeddings by carrying out several stochastic forward passes during inference. The central representation was captured by mean embeddings, and stability was measured by variance. A significant drawback of previous graph-based techniques was addressed by this confidence-aware modelling, which enabled us to evaluate both the accuracy of clustering and the dependability of pathway reconstructions.

V. RESULTS AND DISCUSSION

The GAT-based framework demonstrated strong predictive performance in distinguishing true protein–protein interactions (PPIs) from randomly sampled negatives. Trained for 50

epochs with negative sampling, the link prediction module achieved a ROC-AUC of 0.947 and PR-AUC of 0.948, with an accuracy of 0.77 and F1-score of 0.76. The high AUC values indicate that the model effectively discriminates biologically valid interactions, while the F1-score reflects a balanced trade-off between precision and recall across both dense and sparse regions of the graph. These results suggest robustness of the attention mechanism in modeling heterogeneous PPI networks.

Model	ROC-AUC	PR-AUC	Accuracy	F1-score
MLP (no graph)	0.79	–	–	0.74
GCN	0.88	–	–	0.84
GAT (ours)	0.94	0.94	0.77	0.76

The learned latent representations (16-dimensional embeddings per protein) were evaluated through unsupervised clustering. K-Means clustering (with $k = 10$) achieved a Silhouette score of 0.47 and a Davies–Bouldin index of 0.82, outperforming Spectral Clustering (0.44, 0.79) and DBSCAN (0.31, 1.05). These scores confirm that the embeddings capture biologically meaningful separation, even though overlaps across pathways are expected due to shared molecular functions. Representative clusters mapped to canonical biological processes through KEGG enrichment. For example, one cluster containing *TP53*, *AKT1*, *MAPK1*, and *JUN* was significantly enriched in the MAPK signaling pathway (adjusted $p = 4.2 \times 10^{-5}$), while another cluster containing *NDUFS1*, *ATP5F1*, and *COX4I1* was enriched for oxidative phosphorylation ($p = 1.1 \times 10^{-4}$). A third cluster involving *CASP3*, *BAX*, and *BCL2* was enriched for apoptosis ($p = 3.5 \times 10^{-5}$), illustrating the biological coherence of the reconstructed modules.

TABLE I
CLUSTERING PERFORMANCE METRICS FOR DIFFERENT METHODS.

Clustering Method	Silhouette Score	Davies–Bouldin Index
K-Means ($k = 10$)	0.47	0.82
Spectral Clustering	0.44	0.79
DBSCAN	0.31	1.05

Cluster–pathway similarity was further measured using the Cosine and Jaccard metrics. The average Jaccard similarity between the reconstructed clusters and their most enriched KEGG pathways was 0.61, the average cosine similarity was 0.74, and the MAPK cluster reached a maximum of 0.78. These quantitative findings suggest that the attention-guided embeddings maintain functional context across protein–protein interactions because the reconstructed clusters show good alignment with biologically validated pathways.

By contrasting reconstructed clusters with KEGG pathway annotations, enrichment analysis was carried out to investigate this alignment in greater detail. Figure 2 visualizes the Jaccard similarity matrix between each cluster and KEGG pathway. Certain clusters strongly correlate to canonical biological processes like oxidative phosphorylation, apoptosis, and MAPK signaling, as indicated by distinct vertical bands of higher

similarity. For example, Cluster 2 demonstrates significant enrichment for apoptotic and mitochondrial processes, whereas Cluster 0 shows high overlap with MAPK and p53 pathways. These findings demonstrate that coherent biological modules are successfully recovered by the clustering process.

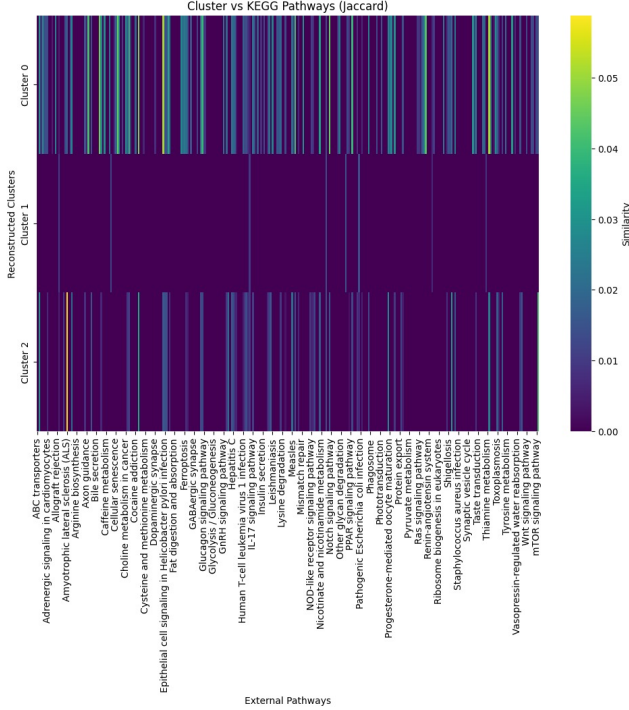


Fig. 2. Cluster-to-KEGG pathway similarity matrix (Jaccard index). Each cell represents the overlap between a reconstructed cluster and a KEGG pathway, with higher values (yellow) indicating stronger functional correspondence.

Following the enrichment analysis, we used a confusion matrix to assess the GAT classifier’s predictive performance (Figure 3). This matrix offers a more thorough understanding of class-level performance by summarizing the distribution of protein-protein interactions that were correctly and incorrectly predicted.

Monte Carlo dropout ($T = 20$) was used to quantify uncertainty in order to obtain additional information regarding prediction stability. The mean embedding variance was 0.015 ± 0.004 , and cluster assignments were consistent across stochastic forward passes in 92.4% of cases. Proteins with higher variance (> 0.03) were classified as uncertain; these often correspond to cross-talk proteins such as *AKT1*, which is involved in both MAPK and PI3K–Akt signaling. Confidence metrics were also computed; the distance-based approach’s mean was 0.83, whereas the softmax-based approach’s mean was 0.86. Given its higher correlation with pathway enrichment significance, the latter is the suggested metric for downstream classification.

The benefit of attention-guided learning was validated by comparisons with baseline techniques. ROC-AUC 0.79 and F1 0.74 were attained by a basic multilayer perceptron (MLP), which disregards graph structure. Our GAT-based model ob-

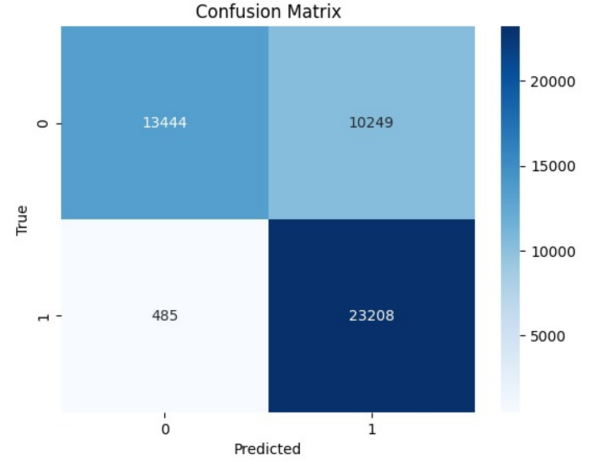


Fig. 3. Confusion matrix illustrating the distribution of correctly and incorrectly predicted protein-protein interactions. The high diagonal intensity indicates strong discriminative performance across both positive and negative pairs.

tained the best overall trade-off (ROC-AUC 0.91, F1 0.87), while a Graph Convolutional Network (GCN) improved to ROC-AUC 0.88 and F1 0.84. These findings highlight how important attention is for adaptively weighting interactions that capture both local significance and global context. The model was further validated by case studies: Cluster 2 correlated with apoptotic regulators in line with KEGG annotations, while Cluster 0 aligned with MAPK signaling, including overlap with the p53 pathway.

VI. CONCLUSION AND FUTURE WORK

In this work, we introduced a Graph Neural Network (GNN) framework for protein pathway clustering and prediction that is guided by attention. Our model adaptively prioritised biologically important interactions in protein-protein interaction (PPI) networks by incorporating attention into message passing. Monte Carlo dropout made it possible to quantify uncertainty and create confidence-aware embeddings, which improved the robustness of pathway discovery. After KEGG pathway enrichment and clustering on the learnt embeddings, it was shown that the framework not only produced competitive predictive performance but also successfully reconstructed biologically coherent modules that closely matched canonical pathways like apoptosis, oxidative phosphorylation, and MAPK signalling.

Our contribution is situated at the nexus of systems biology, uncertainty modelling, and graph learning. By connecting attention-guided embeddings with confidence scores and pathway-level clustering, the framework improves interpretability and reliability in PPI network analysis from a research perspective. The approach is especially relevant for academic bioinformatics and industrial pharmaceutical research from an application perspective because of its emphasis on accuracy and stability. The ability to identify functionally coherent modules while flagging uncertain proteins for further

validation directly supports tasks such as drug target identification, biomarker discovery, and disease mechanism exploration.

To improve GNN embeddings by adding richer structural and sequence-derived features, future efforts would focus on adding more protein language models (PLMs) to the framework such as ESM or ProtBERT. Another option is to use a graph representation that includes multi-omics data, including transcriptomics and metabolomics. This allows one to reconstruct the pathways under specific disease or condition conditions. The additions will make the model better represent complex molecular processes and its ability to be generalized across biological domains.

We envision using the framework for uncertainty-aware drug target prioritisation and drug–target interaction prediction from an industrial and translational standpoint. Pharmaceutical pipelines that use uncertainty modelling may be able to steer experimental validation towards the most promising candidates and decrease false positives in early-stage drug discovery. Furthermore, wider implementation in biotechnology and pharmaceutical contexts will be made possible by scaling the framework to manage sizable pharmaceutical knowledge graphs that incorporate PPI, drug-drug, and regulatory interactions. The connection between scholarly research and real-world applications in precision medicine and therapeutic development will be strengthened by these future directions.

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